

Cisco Packet Tracer Lab Guide

Comprehensive Configuration of VLANs, Switch Port Assignment, Router Interconnections, and Static Routing

Author: Network Engineering Simulation Lab | **Tool:** Cisco Packet Tracer | **Status:** Completed & Verified

Task 1: Creating VLANs ('Sales' and 'Support')

Virtual Local Area Networks (VLANs) segment physical Local Area Networks (LANs) into logical broadcast domains. In a simulation environment like Cisco Packet Tracer, creating VLANs allows administrators to isolate traffic, enhance security, and improve network performance.

To configure VLAN 10 ('Sales') and VLAN 20 ('Support') on a Cisco Catalyst 2960 switch, access the Command Line Interface (CLI) in privileged EXEC mode, enter global configuration mode, and define the VLAN IDs and names.

```
Switch> enable
Switch# configure terminal
Switch(config)# vlan 10
Switch(config-vlan)# name Sales
Switch(config-vlan)# vlan 20
Switch(config-vlan)# name Support
Switch(config-vlan)# end
```

Note: VLAN database creation in modern Cisco IOS versions is stored in the `vlan.dat` file located in flash memory. Using standard global configuration mode ensures persistence across reloads.

Task 2: Assigning Switch Ports & Membership Verification

Once VLANs are defined, switch ports must be assigned to their respective VLANs in access mode. To assign three ports to VLAN 10 ('Sales') and two ports to VLAN 20 ('Support') efficiently, the interface range command is utilized.

Port Assignment Configuration

```
Switch# configure terminal
Switch(config)# interface range FastEthernet 0/1 - 3
Switch(config-if-range)# switchport mode access
```

```
Switch(config-if-range)# switchport access vlan 10
Switch(config-if-range)# exit
Switch(config)# interface range FastEthernet 0/4 - 5
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 20
Switch(config-if-range)# end
```

VLAN Membership Verification

To verify that ports are correctly assigned, use the `show vlan brief` command:

```
Switch# show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/6, Fa0/7, Fa0/8, Fa0/9 Fa0/10, Fa0/11, Fa0/12, Fa0/13 Fa0/14, Fa0/15, Fa0/16, Fa0/17 Fa0/18, Fa0/19, Fa0/20, Fa0/21 Fa0/22, Fa0/23, Fa0/24, Gig0/1 Gig0/2
10	Sales	active	Fa0/1, Fa0/2, Fa0/3
20	Support	active	Fa0/4, Fa0/5
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddi-default	active	
1005	trnet-default	active	

Task 3: Router Interconnection & Static Routing Configuration

To connect two routers (Router 1 and Router 2) in Cisco Packet Tracer, establish a serial or Ethernet link between them, assign IP addresses to their respective interfaces, and enable them using the `no shutdown` command.

1. IP Address Configuration

Router	Interface	IP Address	Subnet Mask
Router 1	GigabitEthernet 0/0 (LAN)	192.168.10.1	255.255.255.0
Router 1	Serial 0/1/0 (WAN Link)	10.0.0.1	255.255.255.252
Router 2	Serial 0/1/0 (WAN Link)	10.0.0.2	255.255.255.252
Router 2	GigabitEthernet 0/0 (Remote LAN)	192.168.20.1	255.255.255.0

2. Configuring Static Route on Router 1

To enable Router 1 to reach the remote LAN network behind Router 2 (192.168.20.0/24), configure a static route pointing to Router 2's serial interface IP as the next-hop address.

```
Router1> enable
Router1# configure terminal
Router1(config)# ip route 192.168.20.0 255.255.255.0 10.0.0.2
Router1(config)# end
Router1# copy running-config startup-config
```

Task 4: Inter-VLAN Communication Analysis & Resolution

Why Ping Fails Between 'Sales' (VLAN 10) and 'Support' (VLAN 20)

When a device in VLAN 10 ('Sales') attempts to ping a device in VLAN 20 ('Support'), the ping fails. The underlying reasons include:

- **Layer 2 Isolation:** VLANs are distinct broadcast domains. A standard Layer 2 switch only forwards frames within the same VLAN. Frames cannot cross VLAN boundaries without Layer 3 routing assistance.
- **Different Subnets:** Typically, each VLAN is assigned a separate IP subnet (e.g., VLAN 10 on 192.168.10.0/24 and VLAN 20 on 192.168.20.0/24). Devices on different subnets require a Default Gateway (Router or Layer 3 Switch) to route packets between them.

Key Principle: Layer 2 switches examine MAC addresses and forward frames within the same VLAN. They cannot inspect IP headers or route traffic between different VLAN subnets.

Solutions for Inter-VLAN Communication

To enable communication between VLAN 10 and VLAN 20, one of the following architectural approaches must be implemented:

1. Router-on-a-Stick (Inter-VLAN Routing):

Connect an external router to the switch via a single physical link configured as an 802.1Q trunk port. On the router interface, configure subinterfaces corresponding to each VLAN and assign gateway IP addresses using encapsulation dot1q.

```
Router(config)# interface GigabitEthernet 0/0.10
Router(config-subif)# encapsulation dot1Q 10
Router(config-subif)# ip address 192.168.10.1 255.255.255.0
Router(config)# interface GigabitEthernet 0/0.20
Router(config-subif)# encapsulation dot1Q 20
Router(config-subif)# ip address 192.168.20.1 255.255.255.0
```

2. Layer 3 Switch (Multilayer Switch):

Replace or supplement the Layer 2 switch with a Multilayer Switch (such as Cisco Catalyst 3560/3750). Enable IP routing globally and create Switched Virtual Interfaces (SVIs) for each VLAN.

```
Switch(config)# ip routing
Switch(config)# interface vlan 10
Switch(config-if)# ip address 192.168.10.1 255.255.255.0
Switch(config)# interface vlan 20
Switch(config-if)# ip address 192.168.20.1 255.255.255.0
```